
Professional Summary

AI/ML Engineer with 3+ years of experience building and deploying ML, deep learning, and Gen AI solutions across financial services and enterprise environments. Expertise in Transformer-based NLP, LLM fine-tuning, RAG systems, and multi-agent AI frameworks, with hands-on experience in fraud detection, risk prediction, and compliance automation using PyTorch, TensorFlow and LangChain.

Technical Skills:

- **Programming & Data:** Python, SQL, PySpark, Pandas; data wrangling, feature engineering, statistical analysis, ETL pipelines
- **Machine Learning & Deep Learning:** Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Keras; CNNs, RNNs, LSTMs, Transformers; supervised/unsupervised modeling, hyperparameter tuning, A/B testing
- **NLP & Generative AI:** BERT, GPT, LLaMA, Hugging Face Transformers, LangChain, LlamaIndex, RAG, prompt engineering, RLHF, multi-agent frameworks (LangGraph, AutoGen, Semantic Kernel), sentence-transformers, function calling
- **Cloud & Deployment:** AWS (SageMaker, Bedrock, Lambda, S3, API Gateway), Docker, Kubernetes, Flask APIs, CI/CD pipelines
- **MLOps & Monitoring:** MLflow, AWS CloudWatch, drift detection, model versioning, governance, automated retraining

Work Experience:

JP Morgan & Chase, NY | AI Engineer**Sep 2025 – Present**

- Developed Generative AI, Transformer-based NLP, and ML models for financial risk prediction, fraud detection, compliance monitoring, and Q&A systems using Python, PyTorch, TensorFlow, Scikit-learn, and Hugging Face Transformers.
- Built Retrieval-Augmented Generation (RAG) and document Q&A systems with LangChain, LlamaIndex, Pinecone, and Chroma, integrating transformer embeddings and hybrid search for knowledge retrieval from large-scale financial datasets.
- Designed multi-agent AI workflows leveraging LangGraph, AutoGen, and Semantic Kernel with transformer-based reasoning agents to enable intelligent decision-making, automated report generation, and anomaly detection across financial datasets.
- Conducted data wrangling, feature engineering, and statistical analysis on large transactional datasets using PySpark, SQL, and Pandas, improving Transformer model accuracy and business insights.
- Leveraged AWS (SageMaker, Bedrock, Lambda, S3) for training and deployment of Transformer-based LLMs and agentic AI systems, ensuring compliance with enterprise security standards and scalable financial production pipelines.
- Applied prompt engineering and reinforcement learning with human feedback (RLHF) to optimize LLM contextual accuracy, reduce hallucinations, and enhance Transformer-based financial document interpretation and classification tasks.

Meta, NY | AI/ML Engineer**Mar 2025 – Aug 2025**

- Contributed to a production-grade RAG system for Meta AI Assistant using Python, Hugging Face, FAISS, and LLMs (Llama/GPT), improving response relevance by 30% and reducing hallucinations via hybrid dense + sparse retrieval optimization.
- Developed multilingual NLP models using BERT, XLM-RoBERTa, and Hugging Face tokenizers for intent detection and entity extraction across 50+ languages, deployed on AWS SageMaker, improving query understanding accuracy by 24%.
- Designed and implemented a personalization and memory framework using embedding models (sentence-transformers), vector databases, and SQL-based feature stores, improving contextual ranking and increasing user engagement by 18%.
- Optimized LLM inference performance using quantization (INT8/FP16), ONNX Runtime, DeepSpeed, and GPU batching (CUDA), reducing average latency from 1.6s to 0.85s at production scale while maintaining accuracy.
- Built an LLM tool-use agent system using function calling APIs and orchestration frameworks, enabling search, summarization, and workflow automation, improving task success rate by 26%.
- Engineered large-scale data pipelines with PySpark, SQL, Databricks, AWS Glue, and S3 to process 10B+ conversation logs, supporting finetuning workflows and reducing preprocessing time by 38%.

Accenture | Machine Learning Engineer | Hyderabad, India**May 2022 – Jun 2024**

- Devised deep learning models including CNN and RNN architectures using TensorFlow and PyTorch for automated document classification and fraud transaction detection, increasing operational throughput and reducing manual review efforts.
- Optimized and fine-tuned BERT-base and GPT-2 models via Hugging Face Transformers for customer sentiment analysis and intent detection across support tickets, improving NLP classification accuracy by 20% and reducing customer escalation times.
- Formed reusable PyTorch and TensorFlow training pipelines with automated hyperparameter tuning and batch inference, reducing model training time by 60% and ensuring reproducibility and scalability across deployments.
- Formulated and deployed real-time anomaly detection services using LSTM networks and OpenCV-powered transaction receipts parsing via Flask APIs, reducing fraud monitoring efforts by 55% and strengthening real-time compliance controls.
- Transformed legacy Scikit-learn and XGBoost models to AWS SageMaker using containerized deployments with Docker, establishing API Gateway and Lambda endpoints, reducing deployment time by 50% and enabling seamless auto-scaling.
- Orchestrated model evaluation, logging, and drift detection with MLflow and AWS CloudWatch, enhancing governance, reducing production incidents by 40%, and ensuring continuous model performance and reliability across deployed models.

Education:

- **Masters in management information systems** from Robert Morris University, Moon Township, PA, USA.
- **Bachelors in computer science and engineering** from CMR Institute of Technology, Hyderabad, India.